

Artificial Intelligence & Soft Computing (AISC) Master Viva & Oral Exam Guide

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MODULE 1: AI AGENTS & PEAS FRAMEWORK (EXPERIMENT 1)

Q1: What is an AI Agent? What is an Agent Function vs Agent Program?

- **Agent:** Anything that perceives its environment through **sensors** and acts upon that environment through **actuators**.
- **Agent Function:** Mathematical mapping from percept sequence to action: $f: P^* \rightarrow A$. (Abstract specification).
- **Agent Program:** Concrete software implementation of the agent function executing on a physical architecture.

Q2: What is the PEAS framework? Specify PEAS for an Automated / Self-Driving Car.

- **PEAS:** Performance measure, Environment, Actuators, Sensors.
- **For Automated Car:**
 - **P (Performance):** Passenger safety, speed, legality, comfort, fuel efficiency.
 - **E (Environment):** Roads, vehicles, pedestrians, weather conditions, traffic signals.
 - **A (Actuators):** Steering wheel, accelerator, brakes, horn, turn signals.
 - **S (Sensors):** Video cameras, LiDAR, radar, GPS, speedometer, accelerometer.

Q3: What are the different dimensions of Agent Environments?

1. **Fully Observable vs Partially Observable:** Full access to complete world state at any time vs incomplete/noisy sensor data.
2. **Deterministic vs Stochastic:** Next state is completely determined by current state and agent action vs random/probabilistic outcomes.
3. **Episodic vs Sequential:** Current decision does not affect future decisions vs current actions alter all future choices.
4. **Static vs Dynamic:** Environment does not change while agent is deliberating vs changes concurrently.
5. **Discrete vs Continuous:** Finite, clearly defined states and time steps vs infinitely continuous states/time.
6. **Single-Agent vs Multi-Agent:** Only one agent operating vs multiple interacting/competing agents.

MODULE 2: STATE SPACE SEARCH & WATER JUG PROBLEM (EXPERIMENT 2)

Q4: Formulate the Water Jug problem as a State Space Search.

- **State Representation:** An ordered pair (x, y) where $x \in [0, J_1]$ is water in Jug 1, and $y \in [0, J_2]$ is water in Jug 2.
- **Initial State:** $(0, 0)$ (both jugs empty).
- **Goal State:** $(d, \cdot)$ or $(\cdot, d)$ where d is the target amount (e.g. 2 gallons).
- **Operators / Production Rules:**
 1. Fill Jug 1: $(x, y) \rightarrow (J_1, y)$
 2. Fill Jug 2: $(x, y) \rightarrow (x, J_2)$
 3. Empty Jug 1: $(x, y) \rightarrow (0, y)$
 4. Empty Jug 2: $(x, y) \rightarrow (x, 0)$
 5. Pour Jug 1 into Jug 2 until full: $(x, y) \rightarrow (x - (J_2 - y), J_2)$
 6. Pour Jug 2 into Jug 1 until full: $(x, y) \rightarrow (J_1, y - (J_1 - x))$
 7. Pour all Jug 1 into Jug 2: $(x, y) \rightarrow (0, x + y)$
 8. Pour all Jug 2 into Jug 1: $(x, y) \rightarrow (x + y, 0)$

Q5: What is the mathematical condition for the Water Jug Problem to be solvable?

- **Theorem:** The problem is solvable if and only if the desired quantity d is a multiple of the greatest common divisor of the capacities:
$$d \in \max(J_1, J_2) \bmod \gcd(J_1, J_2) == 0$$
- **Example:** For $J_1 = 4, J_2 = 3, \gcd(4, 3) = 1$. Since $2 \bmod 1 == 0$, 2 gallons is solvable!

Q6: Why is Breadth-First Search (BFS) preferred over Depth-First Search (DFS) for Water Jug?

- BFS visits nodes level-by-level using a FIFO queue.
- It **guarantees** finding the solution with the minimum number of pours/steps.
- DFS may get trapped in an infinite cycle or follow a long sub-optimal path.

🎮 MODULE 3: GAME PLAYING & MINIMAX ALGORITHM (EXPERIMENT 3)

Q7: What is the Minimax Algorithm? What game characteristics does it require?

- An adversarial game-tree search algorithm for **two-player, zero-sum, deterministic, perfect-information** turn-based games (like Tic-Tac-Toe, Chess).
- **MAX player:** Attempts to maximize the utility score (e.g., $+10$ for win).
- **MIN player:** Attempts to minimize MAX's score (e.g., -10 for loss).
- Backtracks leaf evaluations up the tree by choosing $\max$ at MAX levels and $\min$ at MIN levels.

Q8: What is the Time and Space Complexity of Minimax?

- **Time Complexity:** $O(b^m)$ where b is the branching factor and m is maximum tree depth.
- **Space Complexity:** $O(bm)$ (using Depth-First Search).

Q9: What is Alpha-Beta Pruning? Under what mathematical condition is a branch pruned?

- An optimization technique that eliminates subtrees without altering the final minimax decision.
- **α (Alpha):** The best (maximum) value MAX is guaranteed so far (initially $-\infty$).
- **β (Beta):** The best (minimum) value MIN is guaranteed so far (initially $+\infty$).
- **Pruning Condition:** Prune when $\beta \leq \alpha$. Any further exploration cannot change the ancestor's minimax choice.
- **Optimal Time Complexity:** With ideal move ordering, time complexity reduces from $O(b^m)$ to $O(b^{m/2})$.

🔍 MODULE 4: INFORMED HEURISTIC SEARCH & A* SEARCH (EXPERIMENT 5)

Q10: What is the difference between Uninformed and Informed Search?

- **Uninformed (Blind) Search:** Has no information about how close a state is to the goal (e.g., BFS, DFS). Explores systematically.
- **Informed (Heuristic) Search:** Uses domain-specific heuristic knowledge $h(n)$ to estimate remaining distance to the goal, drastically reducing state exploration.

Q11: Explain the A* Search evaluation function: $f(n) = g(n) + h(n)$.

- **$g(n)$:** The exact cost accumulated from the start node to current node n .
- **$h(n)$:** The estimated (heuristic) cost from node n to the goal.
- **$f(n)$:** Total estimated cost of the cheapest solution path through node n .
- A* expands the node with the lowest $f(n)$ value from the Priority Queue (Open List).

Q12: What is an Admissible Heuristic? Why is it mandatory in A*?

- **Definition:** A heuristic is admissible if it **never overestimates** the true cost to reach the goal:
$$h(n) \leq h^*(n) \quad \text{for all } n$$

where $h^*(n)$ is the true optimal cost.
- **Significance:** Admissibility guarantees that A* tree search is **complete and optimal** (always returns the shortest path).

Q13: What is a Consistent (Monotonic) Heuristic?

- **Definition:** A heuristic is consistent if for every node n and every successor n' generated by action a :
$$h(n) \leq c(n, a, n') + h(n')$$

(satisfies the triangle inequality).

- Consistency implies admissibility. With consistency, the first time A^* visits a state, it is guaranteed to be on the optimal path.

Q14: How does A^* behave when $h(n) = 0$ or $g(n) = 0$?

- If $h(n) = 0$: $f(n) = g(n)$ \implies Becomes **Dijkstra's Algorithm / Uniform Cost Search (UCS)**.
- If $g(n) = 0$: $f(n) = h(n)$ \implies Becomes **Greedy Best-First Search** (fast, but sub-optimal).

MODULE 5: LOGIC PROGRAMMING & PROLOG (EXPERIMENT 6)

Q15: What is PROLOG? What logic paradigm does it use?

- PROLOG (Programming in Logic) is a declarative logic programming language.
- Based on **First-Order Predicate Calculus** restricted to **Horn Clauses** (clauses with at most one positive literal).
- Instead of writing step-by-step algorithms, programmer declares **Facts** and **Rules**, and asks **Queries**.

Q16: How does the PROLOG execution engine work?

- Uses **Resolution Refutation** with **Unification** and **Backtracking** (Depth-First Search).
- **Unification**: The algorithmic process of finding a set of variable bindings that makes two terms identical.

Q17: What is the Cut operator (!) in PROLOG? Distinguish Green Cut vs Red Cut.

- **Cut (!)**: A built-in predicate that commits the system to all choices made since the parent goal was invoked, preventing backtracking.
- **Green Cut**: Only improves performance without changing the set of answers/solutions returned.
- **Red Cut**: Changes the declarative meaning and the solutions produced by the program.

MODULE 6: NEURAL NETWORKS & PERCEPTRON (EXPERIMENT 10)

Q18: What is the McCulloch-Pitts Neuron Model?

- The earliest computational model of a biological neuron (1943).
- Has binary inputs x_i in $\{0, 1\}$, fixed excitatory weights $+1$, and inhibitory inputs.
- Fires (1) if sum of excitatory inputs $\geq \theta$ (threshold) and no inhibitory input is active.

Q19: Explain the Single-Layer Perceptron (SLP) and its Learning Rule.

- Introduced by Frank Rosenblatt (1958). Has learnable weights w_i and a bias b .
- Net input: $z = \sum w_i x_i + b$. Output: $y = 1$ if $z \geq 0$, else 0 .
- **Perceptron Learning Rule**:
$$w_i \leftarrow w_i + \eta (y_{\text{target}} - y_{\text{actual}}) x_i$$
where η is the learning rate ($0 < \eta \leq 1$).

Q20: What is Linear Separability? Why does a Single-Layer Perceptron fail on XOR?

- **Linear Separability**: Data classes can be separated by a single straight line (in 2D) or hyperplane (in n D).
- **AND and OR** are linearly separable.
- **XOR (Exclusive-OR)** is NOT linearly separable: points $(0,1)$ and $(1,0)$ require class 1, while $(0,0)$ and $(1,1)$ require class 0. No single line can separate them! Hence SLP fails on XOR (Minsky & Papert, 1969).

Q21: How does a Multi-Layer Perceptron (MLP) solve the XOR problem?

- MLP introduces one or more **Hidden Layers** with non-linear activation functions (Sigmoid, ReLU).
- The hidden layer maps the non-linearly separable inputs into a transformed feature space where they become linearly separable.
- Trained using the **Backpropagation Algorithm** (Gradient Descent with chain rule).

MODULE 7: FUZZY LOGIC & FUZZY SETS (EXPERIMENT 7)

Q22: What is the fundamental difference between Crisp Sets and Fuzzy Sets?

- **Crisp Set:** Binary boundary. An element either belongs or does not belong: $\mu_A(x) \in \{0, 1\}$.
- **Fuzzy Set:** Gradual boundary. An element has a degree of membership between 0 and 1: $\mu_A(x) \in [0, 1]$.

Q23: Define standard Fuzzy Set Operations (Union, Intersection, Complement).

- **Fuzzy Union ($A \cup B$):** Maximum membership:
$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$
- **Fuzzy Intersection ($A \cap B$):** Minimum membership:
$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$
- **Fuzzy Complement ($\bar{A}$):** Difference from 1:
$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

Q24: What is Fuzzy Max-Min Composition?

- Given fuzzy relations $R(X, Y)$ and $S(Y, Z)$, their composition $T = R \circ S$ is:
$$\mu_T(x, z) = \max_{y \in Y} \left(\min(\mu_R(x, y), \mu_S(y, z)) \right)$$

Q25: What is Defuzzification? Explain the Centroid Method.

- **Defuzzification:** The mathematical process of converting a fuzzy set / membership distribution into a single crisp scalar value.
- **Centroid Method (Center of Gravity / COG):** Finds the balance point of the fuzzy area:
$$z^* = \frac{\sum z \cdot \mu(z)}{\sum \mu(z)} \quad \left(\text{or } \frac{\int z \cdot \mu(z) dz}{\int \mu(z) dz} \right)$$

Most widely used defuzzification method due to intuitive physical meaning.